

# [i]NSTANT

## Loops

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- Problem Solving Instructor



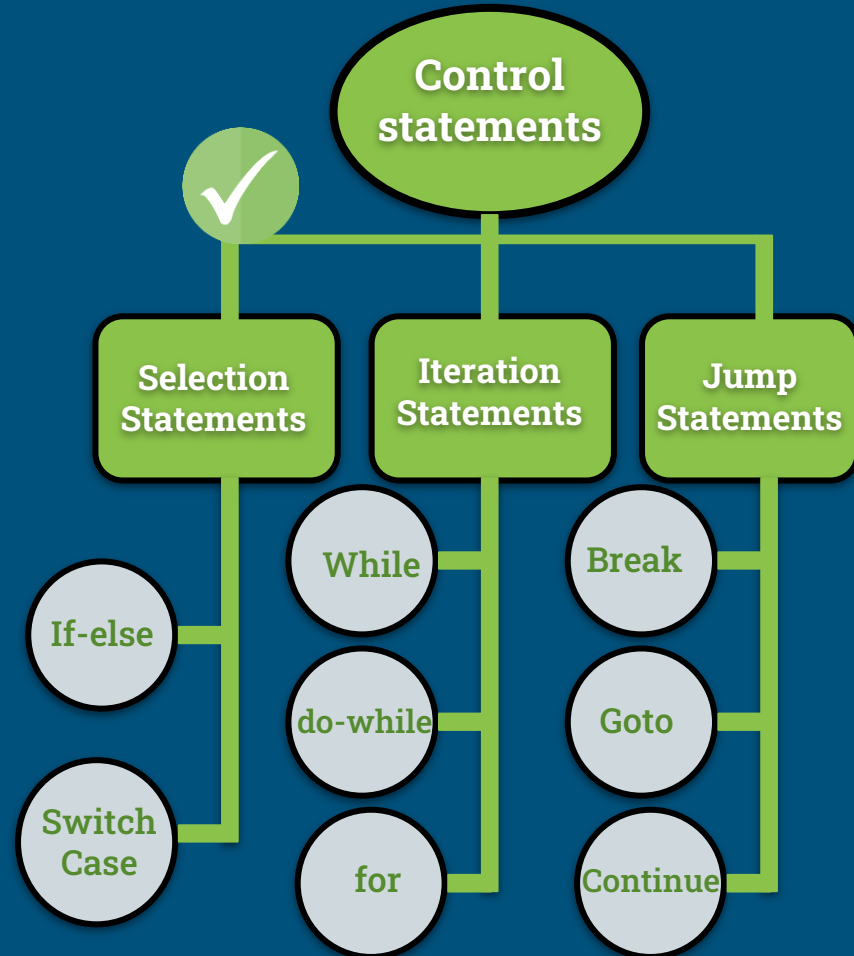
# Agenda

- **Control Statement**
  - **Repetition Statement Types**
    - While Loop
    - For Loop
    - Do While Loop
  - **Jumping Statement**
    - Named Loop ( Label Loop )
    - Goto
    - Continue
    - Break
- **Nested Loop**
- **Bonus Question**

# What is Control Statement ?

In C++ statements in a program are executed in sequential order (From top to bottom)

- This is called sequential execution.
- There are control statements that allow you to choose what code you want to run based on conditions.
- Those control statements are split into two categories : Selection statements and Repetition statements
- We will learn about Selection statements today



# Question

**What If We Want To Print The Numbers From 1 To 5 ?**

```
Cout << 1 << endl ;
```

```
Cout << 2 << endl ;
```

```
Cout << 3 << endl ;
```

```
Cout << 4 << endl ;
```

```
Cout << 5 << endl ;
```

**But What If We Want To Print From 1 To 1000 ?**



# Loops

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## What Is Loops :

In computer Programming, a Loop is used to execute a group of instructions or a block of code multiple times, without writing it repeatedly.

The block of code is executed based on a certain condition

## Loops Types :

- Entry Controlled
  - Exit Controlled
-

# Loops

```
graph TD; Loops --> EntryControlled[Entry Controlled]; Loops --> ExitControlled[Exit Controlled]; EntryControlled --> ForLoop[For - Loop]; EntryControlled --> WhileLoop[While - Loop]; ExitControlled --> DoWhileLoop[Do-while - Loop];
```

Entry Controlled

```
For (initialization ; condition ; updation)
{
}
```

For - Loop

```
while (condition)
{
}
```

While - Loop

Exit Controlled

```
do
{
} while (condition)
```

Do-while - Loop

# Loops

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**To Create Loop You Must Have Two Things :**

1. What We Need To Do ( Statement )
2. How Many Times OR Till How Long Will We Repeat It  
( Expression or Condition )

**In C++ There Is 3 Kinds Of Loops :**

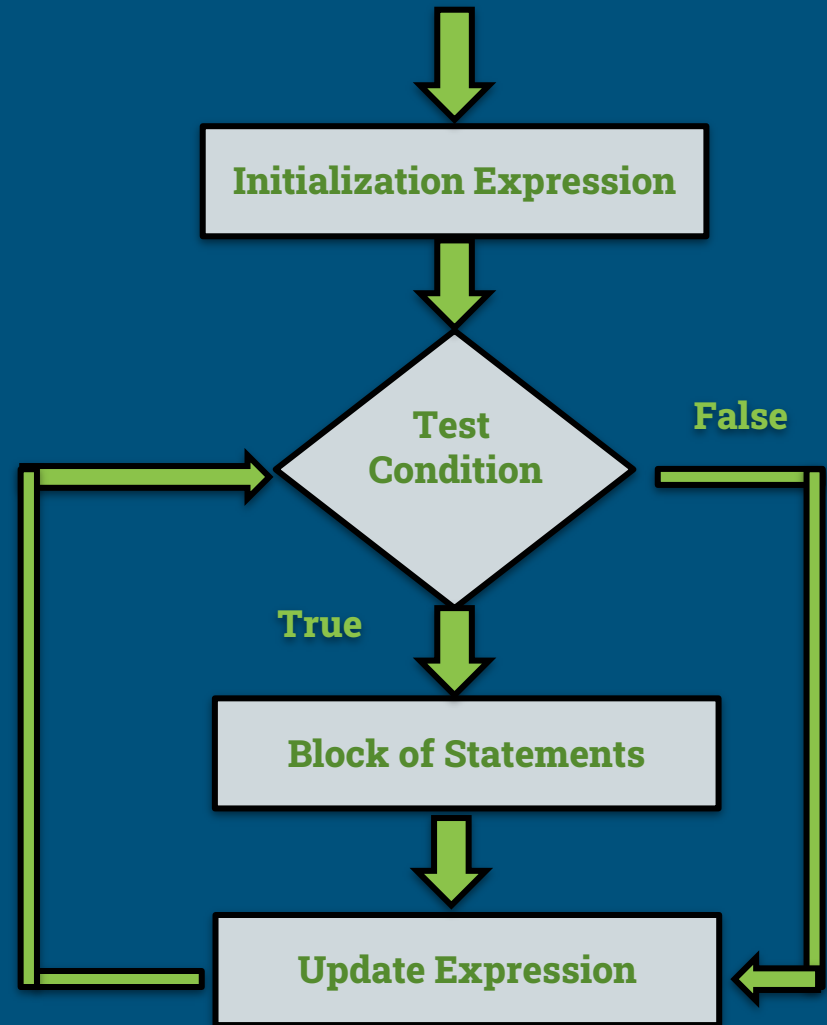
- ❖ For
  - ❖ While
  - ❖ Do-While
-



# For Loop

A For loop is a repetition control structure that allows us to write a loop that is executed a specific number of times.

The loop enables us to perform n number of steps together in one line.







# For Loop

---

```
for (int i = 0 ; i < 10 ; i++){  
    cout << "Yarb A5od E3fa" <<endl;  
}
```

The Code

```
Yarb A5od E3fa  
Yarb A5od E3fa  
Yarb A5od E3fa  
Yarb A5od E3fa  
Yarb A5od E3fa  
Yarb A5od E3fa  
Yarb A5od E3fa  
Yarb A5od E3fa  
Yarb A5od E3fa  
Yarb A5od E3fa
```

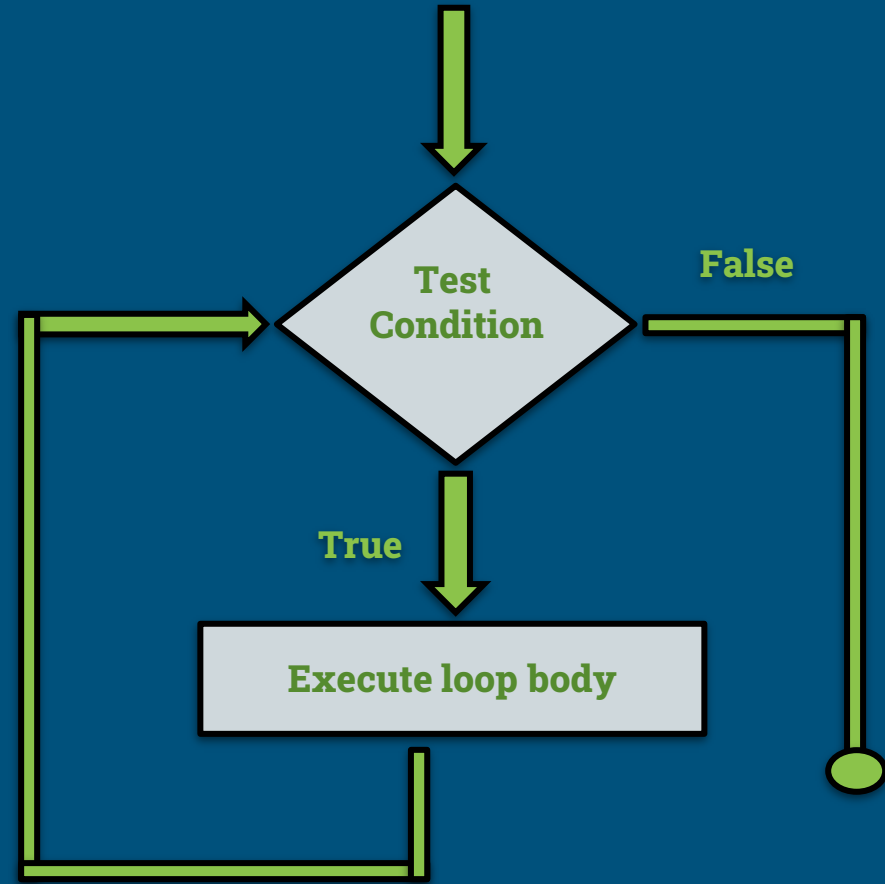
OutPut



# While Loop

while loops are used in situations where we do not know the exact number of iterations of the loop beforehand.

The loop execution is terminated on the basis of the test conditions.





# While Loop

---

```
int t ;
cin >> t ;
while (t > 0){
    cout << "Yarb Mazen Ya5od E3fa" <<endl;
    t-- ;
}
```

The Code

```
10
Yarb Mazen Ya5od E3fa
Yarb Mazen Ya5od E3fa
Yarb Mazen Ya5od E3fa
Yarb Mazen Ya5od E3fa
Yarb Mazen Ya5od E3fa
Yarb Mazen Ya5od E3fa
Yarb Mazen Ya5od E3fa
Yarb Mazen Ya5od E3fa
Yarb Mazen Ya5od E3fa
Yarb Mazen Ya5od E3fa
```

OutPut

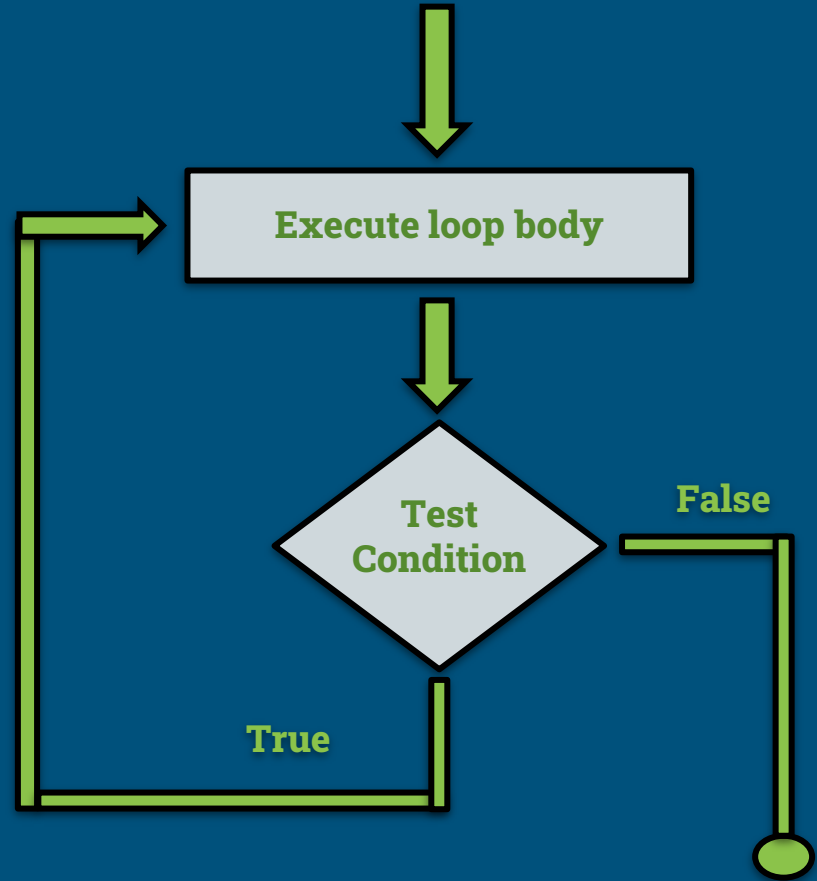


# Do-While Loop

The main difference between a do-while loop and the while loop is in the do-while loop the condition is tested at the end of the loop body

## Note

In a do-while loop, the loop body will execute at least once irrespective of the test condition.





# Do-While Loop



```
do {  
    cout << " At Least Do me Once and Dont  
Forget That In The End We Put Semi Collen " << endl;  
} while (1 > 2);
```

The Code



```
At Least Do me Once and Dont Forget  
That In The End We Put Semi Collen
```

OutPut

# Named Loop ( Label Loop )

The continue statement breaks one iteration (in the loop), if a specified condition occurs, and continues with the next iteration in the loop.

Goto statement helps in transferring the execution of the program from one line to another.

The break in C++ is a loop control statement that is used to terminate the loop.

## Jump Statements

Continue

Goto

Break



# Goto

```
ineligible:
    cout<<"You are not eligible to vote!\n";
    cout<<"Enter your age:\n";
    int age;
    cin>>age;
    if (age < 18){
        goto ineligible;
    }
    else
    {
        cout<<"You are eligible to vote!";
    }
```

The Code

```
You are not eligible to vote!
Enter your age:
16
You are not eligible to vote!
Enter your age:
17
You are not eligible to vote!
Enter your age:
15
You are not eligible to vote!
Enter your age:
19
You are eligible to vote!
```

OutPut



# Break

```
int x = 0 ;  
while(true){  
    x++ ;  
    if (x == 5){  
        cout << "We Will Break Here" <<endl;  
        break;  
    }  
    cout << x <<endl;  
}
```

The Code

```
1  
2  
3  
4  
We Will Break Here
```

OutPut





# Continue

```
for (int i = 1 ; i <= 10 ; i++){  
    if (i == 5){  
        continue;  
    }  
    cout << i <<endl;  
}
```

The Code

```
1  
2  
3  
4  
6  
7  
8  
9  
10  
Process finished with exit code 0
```

OutPut

# Nested loop

**Nested loop means a loop statement inside another loop statement.**

**That is why nested loops are also called as “ loop inside loop “.**

*Note*

There is no rule that a loop must be nested inside its own type. In fact, there can be any type of loop nested inside any type and to any level.



# Nested loop

## Syntax for Nested For loop .



```
#include <iostream>
using namespace std;

int main() {
    for ( initialization; condition; increment ) {

        for ( initialization; condition; increment ) {

            // statement of inside loop
        }

        // statement of outer loop
    }
}
```



# Nested loop



## Syntax for Nested While loop

```
● ● ●  
  
#include <iostream>  
using namespace std;  
  
int main() {  
    while(condition) {  
  
        while(condition) {  
  
            // statement of inside loop  
        }  
        // statement of outer loop  
    }  
}
```



# Nested loop



## Syntax for Nested Do-While loop

```
● ● ●  
  
#include <iostream>  
using namespace std;  
  
int main() {  
    do{  
  
        do{  
  
            // statement of inside loop  
        } while(condition);  
  
        // statement of outer loop  
    } while(condition);  
}
```

## Task Question

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- **Difference Between return & and Break**



# Bonus Question

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- **Problem Statement**
- Word Counting
- Little Bear Want Be Big
- Make Groups
- Masterpiece Problem





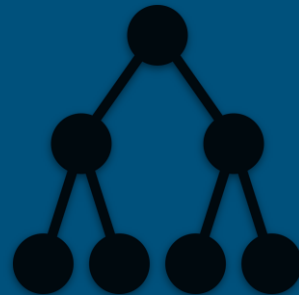
*Any Questions*







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Thank You



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